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MULTI-COMPONENT AUXILIARY POWER SYSTEM AND WORK VEHICLE

Abstract

A multi-component auxiliary power system includes a power transfer linkage configured to be coupled to a drive shaft to transmit rotational power from a prime mover to the power transfer linkage. The power transfer linkage includes a first rotational power output and a second rotational power output. The system further includes a plurality of different auxiliary power components that are operatively coupled to the power transfer linkage. The power transfer linkage transfers and distributes rotational power from the drive shaft to each of the plurality of auxiliary power components. The plurality of different auxiliary power components comprises a hydraulic pump attached to the power transfer linkage and an air compressor attached to the power transfer linkage.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application claims priority to U.S. Provisional Patent Application Ser. No. 63/560,225 filed on Mar. 1, 2024, the disclosure of which is incorporated herein by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The invention generally relates to multi-component auxiliary power systems and to work vehicles including such a multi-component auxiliary power system.

Description of Related Art

[0003] Welder systems for electrical welding, such as arc welding or MIG welding require some electrical power source to provide the electrical power necessary to drive the welding process. Typical welder systems obtain the necessary electrical power directly from an AC or DC power source that has a relatively stable wave form, which requires relatively simple circuitry to convert the electrical power received from the power source into the steady wave form needed for the welding process. Such systems are typically limited to very specific functionalities and used at particular locations due to the specific electrical power source needed. However, welders are often used in field environments where specific electrical power sources may be difficult to access and/or lack certain functionalities. Therefore, welding systems are desired that provide additional functionality that may be needed in a field environment and/or are less restricted geographically by access to a specific AC or DC electrical power source.

[0004] To provide the added field portability, welding systems are sometimes carried on a work truck. Work trucks are primarily used to perform jobsite work functions. The truck as it comes delivered from the vehicle factory is simply configured to transport people and goods. To provide for more specialized field applications, it is common to modify the basic work truck to convert it from a simple transport vehicle into a specialized work truck. Such converted work trucks are often modified by the integration of additional machinery components, such as generators, welders, and hydraulic cranes onto the truck's frame and/or power and/or control systems. The optimization of components on work trucks typically revolves around several key objectives, including weight reduction, space efficiency, and enhanced uptime, as well as lengthening and/or streamlining maintenance cycles. Prioritizing these goals typically involves minimizing the overall weight of the truck, ensuring efficient space utilization, and increasing operational uptime. Effectively designing to these measures for a specific work truck design can collectively contribute to a more versatile, resource-efficient, and reliable work truck system.

[0005] Often, a specialized work truck is equipped with a transmission designed with a power take-off port, which provides a gateway for coupling a gearbox called a power take-off (PTO) to provide a rotational power source that can be coupled with other auxiliary systems via a rotational drive shaft to provide kinetic power to the auxiliary systems. The PTO system enables the direct driving of various machines and/or shafts, providing additional versatility and functionality to standard work trucks. However, PTOs and transmission PTO ports operate within strict constraints of space, torque, angular velocity (e.g., RPM), and power, which can limit the flexibility of their use with regard to the different types and/or number of machines that can draw driving power from the PTO system.

[0006] In view of the above, it would be desirable to have a system that can use the energy from a single PTO drive of a work truck and use it for powering a wider variety of different types of

machines without having to make significant changes to the system when switching between and/or simultaneously using machines that require different forms of power, such as compressed air power, hydraulic power, and/or electric power.

BRIEF SUMMARY OF THE INVENTION

[0007] The following summary presents a simplified summary in order to provide a basic understanding of some aspects of the devices, systems and/or methods discussed herein. This summary is not an extensive overview of the devices, systems and/or methods discussed herein. It is not intended to identify critical elements or to delineate the scope of such devices, systems and/or methods. Its sole purpose is to present some concepts in a simplified form as a prelude to the more detailed description that is presented later.

[0008] In accordance with one aspect of the present invention, provided is a multi-component auxiliary power system. The system comprises a power transfer linkage configured to be coupled to a drive shaft to transmit rotational power from a prime mover to the power transfer linkage. The power transfer linkage includes a first rotational power output and a second rotational power output. The system further includes a plurality of different auxiliary power components that are operatively coupled to the power transfer linkage. The power transfer linkage transfers and distributes rotational power from the drive shaft to each of the plurality of auxiliary power components. The plurality of different auxiliary power components comprises a hydraulic pump attached to the power transfer linkage and an air compressor attached to the power transfer linkage.

[0009] In accordance with another aspect of the present invention, provided is a multi-component auxiliary power system. The system includes a power transfer linkage configured to be coupled to a rotational power coupling to transmit rotational power from a prime mover to the power transfer linkage. The power transfer linkage includes a first rotational power output and a second rotational power output. The system further includes a plurality of auxiliary power components operatively coupled to the power transfer linkage. The power transfer linkage transfers rotational power from the prime mover to each of the plurality of auxiliary power components. The plurality of auxiliary power components includes an air compressor attached to the power transfer linkage and driven by the first rotational power output of the power transfer linkage, an electric generator attached to the air compressor and driven by a through-shaft of the air compressor, and a hydraulic pump attached to the power transfer linkage and driven by the second rotational power output of the power transfer linkage.

[0010] In accordance with another aspect of the present invention, provided is a vehicle including a frame supported by one or more wheels, an engine supported by the frame, a transmission driven by the engine, and a multi-component auxiliary power system supported by the frame. The multi-component auxiliary power system includes a power transfer linkage coupled to a rotational power coupling to transmit rotational power from the engine to the power transfer linkage. The power transfer linkage includes a first rotational power output and a second rotational power output. The multi-component auxiliary power system further includes a plurality of auxiliary power components operatively coupled to the power transfer linkage. The power transfer linkage transfers rotational power from the engine to each of the plurality of auxiliary power components. The plurality of auxiliary power components includes an air compressor attached to the power transfer linkage and driven by the first rotational power output of the power transfer linkage, an electric generator attached to the air compressor and driven by a shaft of the air compressor, and a hydraulic pump attached to the power transfer linkage and driven by the second rotational power output of the power transfer linkage.

[0011] According to a nonlimiting aspect of the invention, a multi-component auxiliary power system includes a power transfer linkage, which may be coupled to a rotational power coupling to transmit rotational power from a prime mover to the power transfer linkage. A plurality of different auxiliary power components may be operatively coupled to the power transfer linkage. The power transfer linkage may transfer and distribute rotational power, such as from the prime mover,

directly and/or indirectly to each of the plurality of auxiliary power components. At least some of the auxiliary power components may convert the distributed rotational power from the power transfer linkage into a different form of energy.

[0012] According to another nonlimiting aspect of the invention, a work vehicle is provided that includes a frame supported by one or more wheels and a prime mover comprising an engine supported by the frame. A multi-component auxiliary power system as described herein may be supported by the frame. The rotational power coupling may transmit rotational power from the prime mover to the power transfer linkage.

[0013] Technical aspects of multi-component auxiliary power systems as described above preferably include the ability to provide an improved source of multiple forms of power on a jobsite or other remote location. The multi-component auxiliary power system may be provided in the configuration of a PTO-shaft-driven underdeck all-in-one power system for mounting to a work truck to provide multiple sources and/or types of power at a work site from a single power source in the form of the work truck's primary engine and/or PTO system.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] The foregoing and other aspects of the invention will become apparent to those skilled in the art to which the invention relates upon reading the following description with reference to the accompanying drawings, in which:

[0015] FIG. 1 is a perspective view of a multi-component auxiliary power system according to certain non-limiting aspects of the invention;

[0016] FIG. 2 is a side elevation view of the multi-component auxiliary power system;

[0017] FIG. 3 is an exploded perspective view of the multi-component auxiliary power system;

[0018] FIG. 4 is a perspective view of the multi-component auxiliary power system in an illustrative configuration as a PTO-shaft-driven underdeck all-in-one power system connected to a PTO drive and transmission of a work truck (not shown), and

[0019] FIG. 5 shows a vehicle having the multi-component auxiliary power system.

DETAILED DESCRIPTION OF THE INVENTION

[0020] The present invention generally relates to multi-component auxiliary power systems and to work vehicles including such a multi-component auxiliary power system. The present invention will now be described with reference to the drawings, wherein like reference numerals are used to refer to like elements throughout. It is to be appreciated that the various drawings are not necessarily drawn to scale from one figure to another nor inside a given figure, and in particular that the size of the components are arbitrarily drawn for facilitating the understanding of the drawings. In the following description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the present invention. It may be evident, however, that the present invention can be practiced without these specific details. Additionally, other embodiments of the invention are possible and the invention is capable of being practiced and carried out in ways other than as described. The terminology and phraseology used in describing the invention is employed for the purpose of promoting an understanding of the invention and should not be taken as limiting.

[0021] As used herein, “at least one”, “one or more”, and “and/or” are open-ended expressions that are both conjunctive and disjunctive in operation. For example, each of the expressions “at least one of A, B and C”, “at least one of A, B, or C”, “one or more of A, B, and C”, “one or more of A, B, or C” and “A, B, and/or C” means A alone, B alone, C alone, A and B together, A and C together, B and C together, or A, B and C together. Any disjunctive word or phrase presenting two or more alternative terms, whether in the description of embodiments, claims, or drawings, should

be understood to contemplate the possibilities of including one of the terms, either of the terms, or both terms. For example, the phrase “A or B” should be understood to include the possibilities of “A” or “B” or “A and B.”

[0022] In preferred but nonlimiting embodiments, multi-component auxiliary power systems disclosed herein provide one or more mechanically-linked auxiliary power components, as nonlimiting examples, one or more air compressors, electric generators for welding and/or providing battery power, and/or hydraulic pumps, which are mechanically linked with each other by one or more mechanical power transfer mechanisms. Such a power system is configured to receive mechanical energy from a single primary rotary power source and distribute the power to one or more various auxiliary power components, which can transform the energy into a variety of different power types, such as compressed air power, hydraulic power, mechanical kinetic power, and/or electric power, for use by a variety of different types of power machinery. The multi-component auxiliary power system is configured to receive rotary kinetic energy from a primary rotary power source, such as a drive shaft driven by a PTO or engine, or similar rotary drive linkage. The multi-component auxiliary power system may optionally include an electric generator that is powered by a through-shaft drive. The components in the multi-component auxiliary power system may be directly coupled with each other, for example by a direct drive shaft coupling, or coupled with each other via other mechanical power transmission mechanisms, such as gears, belts, clutches, and/or additional shafts. The multi-component auxiliary power system may optionally provide an additional power take-off (PTO) port as yet another auxiliary power component. In some nonlimiting embodiments, the system may be configured to provide compressed air at various cubic feet per minute (CFM) and pressure, weld power electric wave forms, jumpstarting electric waveforms, and battery charging electric waveforms, and/or AC and/or DC export power. In some preferred configurations, the multi-component auxiliary power system may be configured specifically to be incorporated into a work truck or similar automotive vehicle with a PTO driven by the vehicle's primary combustion engine or a secondary engine or other power system. The power system can be provided as a single unit with its components fixedly attached to each other, for example with bolts, welds, threads, etc., configured to be attached to and carried by the frame of a work truck underneath the bed deck of the truck so as to free up deck space for other equipment and/or uses.

[0023] The integration of various power sources is achieved through the mechanical connection of multiple rotating machines (e.g., auxiliary power components). These machines offer the flexibility to operate independently, collaboratively, and/or in a blended manner, all powered by a single prime mover (e.g., the engine and/or PTO from the work truck). This system simplifies safety systems by enabling a centralized power cut-off. The system also typically streamlines control mechanisms for the various auxiliary power components. The common control scheme throughout the mechanically linked system facilitates seamless interaction with the prime mover, allowing for objectives like speed control and auto start-stop functionality. This unified and interconnected approach enhances both the operational efficiency and safety of multi-component auxiliary power systems incorporating principles of the present disclosure.

[0024] Turning now to the nonlimiting embodiments represented in the drawings, FIGS. **1-4** show a multi-component auxiliary power system **10** (hereinafter, sometimes simply referred to as the “power system”) according to an illustrative configuration in accordance with aspects of the invention configured for under deck installation to the frame of a work truck. It is understood, however, that the power system **10** may be installed in other manners and/or on other support configurations and/or other types of use configurations. The power system **10** includes a rotational power coupling, such as a drive shaft **12**, a power transfer linkage **14**, and a plurality of auxiliary power components operatively coupled with the power transfer linkage **14** such that rotational power from the drive shaft **12** is converted by the power transfer linkage **14** and distributed to any one or more of the auxiliary power components. The auxiliary power components of the power

system **10** include a series of mechanically coupled power transforming rotary machines. In the nonlimiting example shown in the drawings, the auxiliary power components include an air compressor **16**, an electric generator **18**, and a hydraulic pump **20** (collectively also referred to herein as the auxiliary power components **16**, **18**, and **20**). The power system **10** may be referred to as a PTO driven compressor, welder, generator, and hydraulic system. The auxiliary power components **16**, **18**, and **20** do not necessarily need to be assembled in the order shown in the drawings, but may be arranged in any possible layout as best meets the space needs of a particular use configuration.

[0025] Preferably, the driving arrangement is such that the input and output speed and torque are congruent to what the mating machine is reliably able to receive and/or transmit. In some foreseeable embodiments, the total summation of the power requirements of all the rotating machines in the power system **10** (e.g., auxiliary power components **16**, **18**, and **20**) may be more than the power output of the mechanical driving source (e.g., the truck engine and/or PTO) driving the drive shaft **12** can support. In this case, the auxiliary power components **16**, **18**, and **20** are preferably configured and controlled in such a fashion as to function within the power limits of the prime mover/primary mechanical driving source, such as the primary or auxiliary engine of a mobile work truck.

[0026] The rotational power coupling between the prime mover and the power transfer linkage **14** may include at least one or more belt drives, gear drives, PTO drives, hydraulic PTO drives, and/or electric PTO drives. In the example shown in FIG. **4**, the rotational power coupling is the drive shaft **12**, which is operatively coupled to the primary mechanical driving source on a mechanical power generation system. For example, the drive shaft **12** is coupled to a PTO port **24** on a work truck, and the mechanical power generation system includes a transmission **22** on the work truck that drives a PTO **26** and the PTO port **24**. In this example, the prime mover (primary mechanical energy source) for the system **10** may be either the primary internal combustion engine for the work truck or an auxiliary engine carried by the work truck. However, the power system **10** can be arranged in other installation configurations, and/or other types of prime movers could be used. For example, the power system **10** may be configured for use with a stationary power plant (e.g., a stationary engine on a stationary platform), the power plant on a railroad locomotive, boat, airplane, trailer, etc. The drive shaft **12** may be single part or multi-part. The drive shaft **12** may include any necessary bracketry and/or support mechanisms to ensure proper fitment, alignment, and longevity of the joints. Although a single driveline is shown, the drive shaft **12** in some embodiments might not be used and other types of rotational power couplings could be used to provide a functional auxiliary power system **10**. For example, the rotational power coupling could include a direct coupling to the transmission **22** or other rotating power source, a soft coupling, a hydraulic coupling, and/or an electrical coupling to the prime mover or mechanical driving source, depending on the type of primary power source to which the power system **10** is being operatively connected.

[0027] The power transfer linkage **14** is configured to transmit the rotational energy from the rotational power supply to one or more different rotationally driven auxiliary power components at transformed power levels and speeds that are appropriate for driving each specific auxiliary power component. The power transfer linkage **14** is coupled to the drive shaft **12** (or other rotational power supply) by any suitable coupling such that the rotational power from the drive shaft **12** is transmitted to the power transfer linkage **14**. For example, the power transfer linkage **14** may include a belt drive and/or a gear drive that functions to direct power through its mechanical linkages to the air compressor **16**, electric generator **18**, and/or hydraulic pump **20**. In this particular nonlimiting example configuration, the power transfer linkage **14** includes a first rotational power output on its rear side, and a second rotational power output on its front side, opposite the rear side. Each of the air compressor **16** and the hydraulic pump **20** is directly connected to different rotational power couplings on the power transfer linkage **14**, and the electric generator **18** is

directly connected to the air compressor. For example, the air compressor **16** is attached to the power transfer linkage **14** and driven by the first rotational power output of the power transfer linkage, and the hydraulic pump **20** is attached to the power transfer linkage and driven by the second rotational power output of the power transfer linkage. Thus, the air compressor **16** can be attached to a first (rear) side of the power transfer linkage **14**, and the hydraulic pump **20** can be attached to a second (front) side of the power transfer linkage that is opposite the first side.

[0028] The power transfer linkage **14** may provide torque reduction, torque increase, and/or rotation directional change from the drive shaft **12** to provide the appropriate transformed power to the respective auxiliary power component. The power transfer linkage **14** can be used to configure the power system **10** to convert the mechanical rotational energy provided by a single power source (e.g., the “prime mover”), such as the primary engine of the work truck, to provide one or more different types of drives, including for example, a belt drive, a PTO drive, a hydraulic PTO drive, and/or an electric PTO drive. In the optional configuration shown in the drawings, the power transfer linkage **14** includes a PTO pad **28** to provide an alternate drive option. The PTO pad **28** can be used to power the hydraulic pump **20** or a drive belt, gearbox, or other rotary equipment. One possible power transfer linkage is a PowerTech San-Gear gearbox, although other types of gear boxes and transmissions could be used.

[0029] The air compressor **16** is driven by a drive coupling on the power transfer linkage **14**. In this example, the air compressor **16** is directly attached to another PTO pad on the rear side of the power transfer linkage opposite the drive shaft **12**, for example with bolts or other fasteners or welds, and coupled to another PTO drive. The air compressor **16** includes a through shaft **30** that in turn provides rotational power to the electric generator **18**. The air compressor **16** may be a rotary screw compressor, piston type compressor, or other type of air compressing machine. The air compressor **16** may be geared or ungeared. In some configurations, the air compressor **16** can be used as a mounting point for the power system **10** to affix to the frame rail of an automobile, such as a work truck, tractor, or construction vehicle, or other framed application, such as a railroad car or locomotive, a truck trailer, a boat, an aircraft frame, or other support structure. In some preferred embodiments, the air compressor **16** is configured to build positive displacement pressure. In one embodiment, the air compressor is a Genair rotary screw air compressor, available from Vanair Manufacturing, Inc.; however other types of air compressors could be used.

[0030] The electric generator **18** includes a dynamo that converts rotational energy into electrical energy. In this example, the electric generator **18** is attached directly to the rear side of the air compressor **16** opposite the power transfer linkage **14**, for example with bolts, screws, and/or welds, and receives power from the through shaft **30** of the electric generator **18**. In other possible configurations, the electric generator **18** may be powered by a belt drive or other type of linkage to the air compressor **16** or from a direct connection to the power transfer linkage **14**. In some nonlimiting embodiments, the electric generator **18** is configured to provide one or both of AC and/or DC electrical power suitable for powering various electrical loads, such as lights, tools, welders, battery chargers and/or jumpers, or other electrical equipment. The electric generator **18** may include any suitable power generation and/or control circuits desired for a particular application. In one example arrangement, the electric generator **18** includes one or more of a welding machine, AC electrical power generation, and/or a battery charging machine. The electric generator **18** can be permanently excited or controlled via external field manipulations.

[0031] The electric generator **18** can be arranged to provide nearly any typical type of electrical power needed. For example, the electric generator **18** may be configured to provide one or both of single-phase or three-phase power, 50 Hz or 60 Hz frequency output, and single- or multi-voltage such as (but not limited to) 120V, 240V, 208V, or 480V. Other output configurations are also possible depending on the needed application.

[0032] The electric generator **18** optionally may include a battery charger system. Preferably such a battery charger system is configured to support multiple charging algorithms that include

algorithms configured to charge lead acid, lithium and/or acting capacitors so that the battery charging function can be used in a variety of voltage source and current source methods. For example, the battery charger system may provide export power using battery charging profiles including lithium chemistry and/or lead acid chemistry. The battery charger system may provide export power that can be used directly for battery charging and/or can be used in conjunction with external power conversion and charging circuits for battery charging.

[0033] The electric generator **18** optionally may optionally include a jump-starting system using starter motor voltage. The jump-starting system preferably provides export power that can be used for directly jumpstarting a vehicle, such as a car, truck, boat, tractor, construction equipment, etc. In some embodiments, the jump-starting system may provide export power that can be used in conjunction with super capacitors and/or other power storage methods or chopper board technology for producing an advanced wave form and voltage manipulation.

[0034] The electric generator **18** optionally may optionally include a welding machine. The welding machine preferably is configured to provide welding wave form manipulation that occurs via field manipulation and or the manipulation of a chopper circuit, which may include a large addition of inductance and capacitance. Of course, other configurations for the electric generator **18** are also possible.

[0035] In the present example, the power transfer linkage **14**, air compressor **16**, electric generator **18**, and hydraulic pump **20** are all fixedly attached with each other, for example with bolts, screws, and/or other types of fasteners, so as to provide a single unit that can be attached to a support, such as the frame of the work truck, at a single location. For example, in the illustrated configuration, the air compressor **16** is configured to be mounted to the frame of the work truck with appropriate fasteners and/or welds, and the remaining components are supported by the air compressor without having to be individually attached to the frame for support. This unitary configuration can provide easier installation and/or removal, for example, for maintenance, and minimizes the need for modifications to the frame of the truck. However, in other configurations, the components could be supported and/or connected with each other directly or indirectly in different arrangements, and the power system **10** is not limited to the exact arrangement, number, or types of components illustrated in the drawings.

[0036] The power system **10** according to some nonlimiting aspects of the present invention can take energy from a single PTO drive of a work truck and use it for powering a wider variety of different types of machines by means of the integration of various forms of rotating power. This innovative approach addresses multiple facets of design optimization discussed above, which typically results in cost reduction, weight optimization, decreased maintenance complexity by reducing multiple prime movers, and/or creation of additional space on the vehicle (e.g., a specialized work truck). This integration can in some configurations both enhance the power efficiency of the work truck and expand its cargo capacity relative to work trucks that are configured according to conventional methods to provide the same various power sources. Moreover, the reduction in fuel consumption during transport of the work truck improves the ecological sustainability and economic benefits of this combination of rotating power sources.

[0037] A work truck for jobsite maintenance, infrastructure, and emergency response purposes (among others) incorporating a power system **10** as disclosed herein has the benefit of being able to provide multiple forms of power on a jobsite. For example, when configured as a PTO shaft driven underdeck power system **10** as shown in FIG. 4, the power system **10** advantageously provides a streamlined system that can replace up to six separate pieces of equipment on a work truck. This configuration can provide additional bed space, enhanced maneuverability, reduced maintenance requirements, freed hitch, and/or an overall reduction in weight of the work truck to which it is attached, which is highly desirable on many job sites. Such configuration can seamlessly integrate beneath the vehicle chassis, opening cargo space while enhancing overall efficiency.

[0038] FIG. 5 shows a vehicle **40**, such as a work truck, having the power system **10** discussed

herein. The vehicle **40** can include a frame **42** or other chassis structure supported by a plurality of wheels **44**. The vehicle has an engine **46** that is coupled to a transmission **22**, and the transmission drives the PTO **26**. In an example embodiment, the air compressor of the power system **10** is directly attached to the frame **42**, and the air compressor supports the power transfer linkage, the hydraulic pump, and the electric generator without such components having to be individually attached to the frame for support. This unitary configuration can provide easier installation and/or removal, for example, for maintenance, and minimizes the need for modifications to the frame **42** of the vehicle.

[0039] It should be evident that this disclosure is by way of example and that various changes may be made by adding, modifying or eliminating details without departing from the fair scope of the teaching contained in this disclosure. The invention is therefore not limited to particular details of this disclosure except to the extent that the following claims are necessarily so limited.

Claims

1. A multi-component auxiliary power system comprising: a power transfer linkage configured to be coupled to a drive shaft to transmit rotational power from a prime mover to the power transfer linkage, wherein the power transfer linkage includes a first rotational power output and a second rotational power output; and a plurality of different auxiliary power components that are operatively coupled to the power transfer linkage, wherein the power transfer linkage transfers and distributes rotational power from the drive shaft to each of the plurality of auxiliary power components; and wherein the plurality of different auxiliary power components comprises a hydraulic pump attached to the power transfer linkage and an air compressor attached to the power transfer linkage.
2. The multi-component auxiliary power system of claim 1, wherein the power transfer linkage includes a first power take-off pad and a second power take-off pad, and further wherein the air compressor is attached to the first power take-off pad and the hydraulic pump is attached to the second power take-off pad.
3. The multi-component auxiliary power system of claim 2, wherein the air compressor is attached to a first side of the power transfer linkage and the hydraulic pump is attached to a second side of the power transfer linkage opposite the first side.
4. The multi-component auxiliary power system of claim 1, wherein the air compressor is configured to generate positive displacement air pressure when driven by the first rotational power output.
5. The multi-component auxiliary power system of claim 1, further comprising an electric generator attached to the air compressor and driven by a through-shaft of the air compressor.
6. The multi-component auxiliary power system of claim 5, wherein the electric generator comprises a battery charger system that generates export power in at least one power profile configured for charging a battery.
7. The multi-component auxiliary power system of claim 6, wherein the at least one power profile is configured for charging a battery with lithium chemistry.
8. The multi-component auxiliary power system of claim 6, wherein the at least one power profile is configured for charging a battery with lead acid chemistry.
9. A multi-component auxiliary power system comprising: a power transfer linkage configured to be coupled to a rotational power coupling to transmit rotational power from a prime mover to the power transfer linkage, wherein the power transfer linkage includes a first rotational power output and a second rotational power output; and a plurality of auxiliary power components operatively coupled to the power transfer linkage, wherein the power transfer linkage transfers rotational power from the prime mover to each of the plurality of auxiliary power components, wherein the plurality of auxiliary power components includes: an air compressor attached to the power transfer linkage

- and driven by the first rotational power output of the power transfer linkage; an electric generator attached to the air compressor and driven by a through-shaft of the air compressor; and a hydraulic pump attached to the power transfer linkage and driven by the second rotational power output of the power transfer linkage.
- 10.** The multi-component auxiliary power system of claim 9, wherein the power transfer linkage includes a first power take-off pad and a second power take-off pad, and further wherein the air compressor is attached to the first power take-off pad and the hydraulic pump is attached to the second power take-off pad.
- 11.** The multi-component auxiliary power system of claim 9, wherein the air compressor is attached to a first side of the power transfer linkage and the hydraulic pump is attached to a second side of the power transfer linkage opposite the first side.
- 12.** The multi-component auxiliary power system of claim 9, wherein the air compressor is configured to generate positive displacement air pressure when driven by the first rotational power output.
- 13.** The multi-component auxiliary power system of claim 9, wherein the electric generator comprises a battery charger system that generates export power in at least one power profile configured for charging a battery.
- 14.** The multi-component auxiliary power system of claim 13, wherein the at least one power profile is configured for charging a battery with lithium chemistry.
- 15.** The multi-component auxiliary power system of claim 13, wherein the at least one power profile is configured for charging a battery with lead acid chemistry.
- 16.** A vehicle, comprising: a frame supported by one or more wheels; an engine supported by the frame; a transmission driven by the engine; and a multi-component auxiliary power system supported by the frame, wherein the multi-component auxiliary power system comprises: a power transfer linkage coupled to a rotational power coupling to transmit rotational power from the engine to the power transfer linkage, wherein the power transfer linkage includes a first rotational power output and a second rotational power output; and a plurality of auxiliary power components operatively coupled to the power transfer linkage, wherein the power transfer linkage transfers rotational power from the engine to each of the plurality of auxiliary power components, and wherein the plurality of auxiliary power components includes: an air compressor attached to the power transfer linkage and driven by the first rotational power output of the power transfer linkage; an electric generator attached to the air compressor and driven by a shaft of the air compressor, and a hydraulic pump attached to the power transfer linkage and driven by the second rotational power output of the power transfer linkage.
- 17.** The vehicle of claim 16, wherein the power transfer linkage includes a first power take-off pad and a second power take-off pad, and further wherein the air compressor is attached to the first power take-off pad and the hydraulic pump is attached to the second power take-off pad.
- 18.** The vehicle of claim 16, wherein the air compressor is attached to a first side of the power transfer linkage and the hydraulic pump is attached to a second side of the power transfer linkage opposite the first side.
- 19.** The vehicle of claim 16, wherein the air compressor is mounted to the frame.
- 20.** The vehicle of claim 16, wherein the electric generator comprises a battery charger system that generates export power in at least one power profile configured for charging a battery.
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